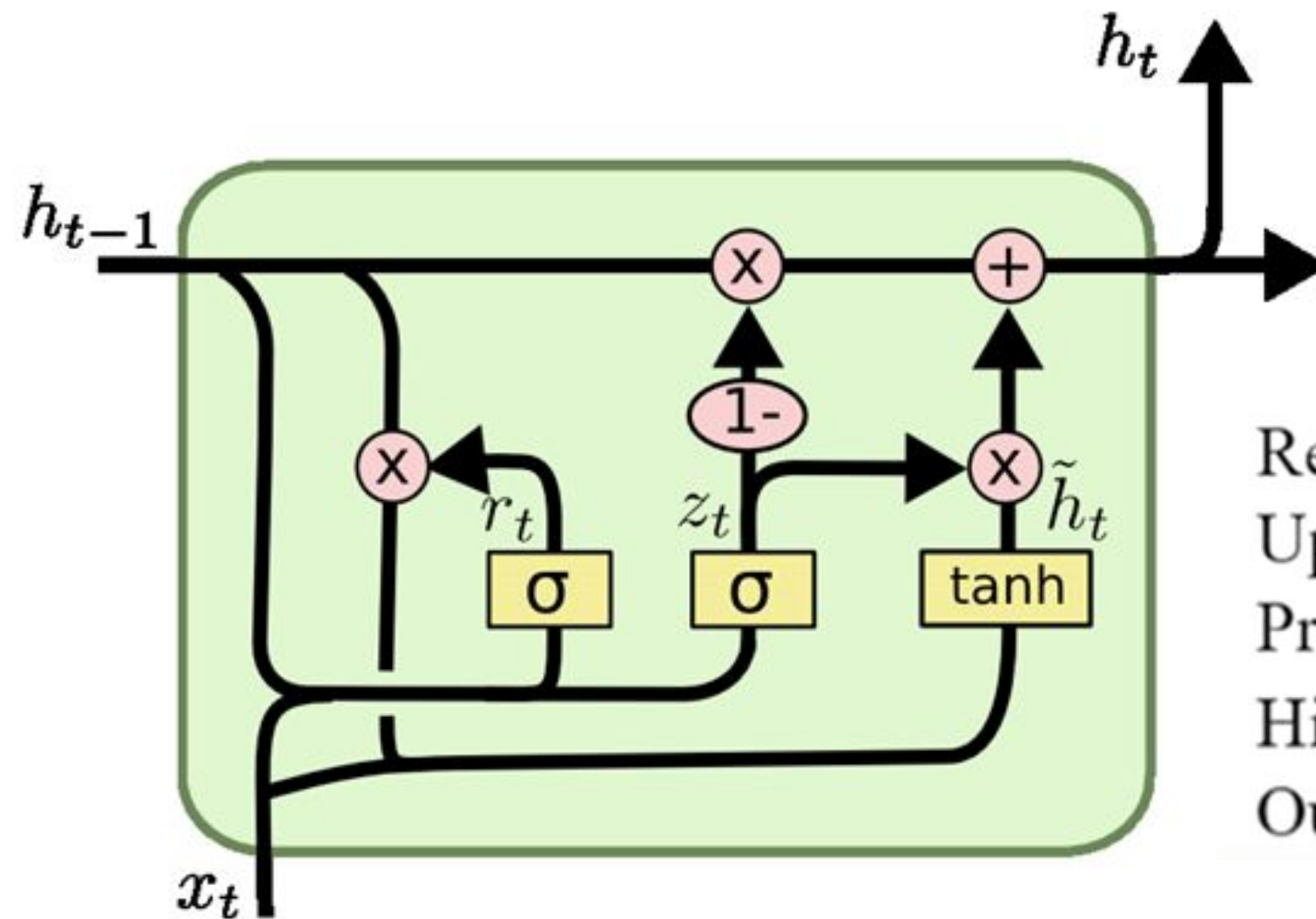


Redes Neuronales Recurrentes (RNNs)

03

Unidades Recurrentes con Compuertas (GRU)

Unidades Recurrentes con Compuertas (GRU)



Reset gate:

Update gate:

Process input:

Hidden state update:

Output:

$$\begin{aligned} r_t &= \sigma (W^{(ir)} x_t + W^{(hr)} h_{t-1}) \\ z_t &= \sigma (W^{(iz)} \bar{x}_t + W^{(hz)} h_{t-1}) \\ \tilde{h}_t &= \tanh (W^{(i\tilde{h})} x_t + W^{(h\tilde{h})} h_{t-1}) \\ h_t &= (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t \\ y_t &= h_t \end{aligned}$$

¡Muchas gracias!